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Wai Yu Amanda Ng

Professional Summary

Data analyst and ML researcher with 3+ years experience building reproducible analytical workflows, modeling high-dimensional data, and designing scalable instructional tools for 400+ learners. Specializes in statistical modeling, and translating complex analyses into actionable insights for education, healthcare, and policy teams data visualizations and stakeholder oriented reports.

Skills

- Experience working within a Linux environment and HPC environment (supercomputer).
- Experiences working in collaborative teams, strong problem-solving and critical thinking skills, attentive to details.
- Languages: English, Cantonese, Mandarin
- Tools: SQL, R, Python (Pandas, NumPy, Matplotlib, PyMC, Scikit-learn, Plotly), Excel, Tableau, Power BI, Git Version Control, QGIS

Education

Sep 2025 – Apr 2026 (expected) **MSc in Statistics**, *University of Toronto*, Toronto, ON, Canada
Courses: Applied Multivariate Analysis, Methods of Applied Statistics, Special Topics in Data Science, Survival Analysis

Sep 2021 – Apr 2025 **H.B.Sc in Statistical Science (Specialist), Mathematics (Major), Health Studies (Focus)**, *University of Toronto*, Toronto, ON, Canada **cGPA: 3.94/4.00**
Courses: Applied Statistical Methods, Applied Spatial Analysis, Time Series Analysis, Applied Bayesian Statistics, Surveys, Sampling and Observational Data, Statistical Consultation
Awards: Samuel Beatty In-course Scholarship (2025), University of Toronto Excellence Award (2024), Eleventh Annual Canadian Statistics Student Conference Undergraduate Level First Prize (2023), University of Toronto Dean's List Scholar (2021–2025)

Professional Experiences

Sep 2025 – Current **R-Based Learning Designer**, *University of Toronto Human Biology Program*

- Built 4 auto-testable R assignments for a cohort of 400 students, cutting manual grading workload by 90% and accelerating feedback turnaround.
- Designed 2 interactive LearnR modules using RShiny, reducing student debugging requests and enhancing self-directed learning efficiency.
- Standardized version-controlled releases on Git across a multi-developer team, delivering zero student-reported issues and significantly improving instructional reliability.

Sep 2024 – Apr 2025 **Data Visualization Analyst**, *University of Toronto Human Biology Program*

- Analyzed learning resource engagement data for 1,000+ course enrollments, identifying behavioral patterns linked to final grades and informing assessment redesign.
- Developed 100+ Tableau dashboards visualizing key performance metrics (discussion board access, assignment completion, early-term quiz grades), enabling instructors to identify at-risk students by week 3.
- Translated analytical findings into data-driven recommendations that improved assessment design and feedback practices, enabling instructors to address early-term performance gaps and refine course delivery pacing.

Jun 2024 – Apr 2025 **Quantitative Research Analyst**, *Centre for Addiction and Mental Health*

- Analyzed how physical movement influenced cybersickness across 2 immersive VR medical educational programs, contributing to 1 peer-reviewed publication advancing VR education research.
- Validated 90+ survey responses and conducted descriptive and non-parametric analyses, identifying a significant positive relationship between movement intensity and nausea severity.
- Delivered data-driven recommendations on movement thresholds and user comfort, informing VR training design decisions to balance educational effectiveness with learner well-being.

May 2023 – Aug 2023 **Data Analyst Intern**, *Canadian Urban Institute*
- Built regression models to analyze nationwide household carbon-emission patterns, identifying key drivers of high-emission regions and guiding targeted interventions for a national sustainable city-planning project.
- Designed an interactive scenario-modeling tool illustrating how urbanization and housing composition influence greenhouse-gas emissions, enabling stakeholders to test policy options on a dashboard.
- Produced 300+ geospatial maps in QGIS and integrated them with statistical findings, uncovering location-specific emission drivers and delivering actionable planning recommendations.

Oct 2022 – May 2023 **Computational Social Science Researcher**, *University of Toronto Population Well-being Lab*
- Applied machine learning methods, Gaussian mixture model, k-means, and hierarchical clustering (mclust, stats) to 150K+ survey observations, uncovering subpopulations based on life satisfaction and demographics.
- Created 4 algorithm tip-sheets explaining statistical methods in accessible terms, improving onboarding and methodological literacy for lab members.
- Designed a poster discussing clustering methods workflow and implications of results, resulting in a conference poster first prize.

Sep 2022 – Dec 2023 **Statistical Consultant**, *University of Toronto Data Sciences Café*
- Provided on-demand statistical guidance to 20+ interdisciplinary researchers, transforming broad research questions into statistically rigorous analysis plans.
- Diagnosed design limitations and recommended tailored analytical approaches, helping researchers avoid common biases and strengthen study validity.
- Authored 5 reports on advanced statistical topics, translating complex concepts and coding methods into clear guidance that enhanced statistical literacy among non-statistician researchers.

Statistics Course Projects

Oct 2025 – Dec 2025 **Uncovering Calendar-Based Structures Through Clustering of Daily Traffic Data**, *STA2005*
- Applied PCA and UMAP to a year-long traffic dataset across 26 locations, uncovering recurring temporal clusters tied to school calendars, holidays, and seasonal routines.
- Created visualizations and prepared a report that informed transportation planning and infrastructure maintenance decisions.

Oct 2025 – Dec 2025 **Fatal US School Shootings (1999-2025)**, *STA4101*
- Modeled fatality risk in 400+ US school shootings using logistic regression, highlighting actionable risk indicators including shooting intent, absence of non-shooter injuries, and shooter connection to the school community.
- Built a reproducible GitHub repository documenting all data cleaning, modeling, and analysis steps to support transparency and collaboration.
- Synthesized findings into a report informing safety and emergency preparedness strategies.

Sep 2025 – Dec 2025 **Statistical Inference for Conditional Spatialtemporal Correlation in Brain MRI Data**, *STA4000*
- Pioneered analysis framework for assessing genetic heritability of intermodal coupling using multi-modal brain MRI data, generating methodological insights for advanced neuroimaging inference.
- Conducted 50+ HPC simulation experiments and applied adaptive clustering techniques with parallel computing to validate proposed model's Type I error control and statistical power.
- Summarized findings in a report that informed methodological refinements for robust neuroimaging statistical approaches.

Feb 2025 – Apr 2025 **Modeling Cocoa Futures Prices: A Statistical Analysis of Ghana's Cocoa Production**, *STA457*
- Integrated 3,700+ daily observations across four datasets, performing extensive cleaning (imputation, correction of invalid entries, aggregation) to produce an analysis-ready dataset.
- Developed ARIMA and Random Forest forecasting models in R (astsa, forcast, randomForest), engineering lagged features and covariates to capture linear and nonlinear drivers of price volatility.
- Conducted full diagnostics (ACF/PACF, Ljung-Box test, residual analysis, feature importance assessment) and authored a comprehensive statistical report on climate and economic influences on commodity pricing.

Oct 2024 – Dec 2024 **Transit and Crime: Investigating Assault Clusters Around Toronto's TTC Subway Lines**, *STA465*
- Analyzed 200k+ Toronto assault incidents using kernel density estimation, point process models and HDB-SCAN clustering to identify assault concentration clusters within varying buffer distances around TTC subway infrastructure.
- Developed interactive geospatial report with maps translating statistical findings into actionable insights for urban planners and policymakers on transit-related public safety.

Sep 2024 – Apr 2025 **Quercus Engagement as a Predictor of First-Year University Student Dropout Risk: A Logistic Regression Analysis**, *STA490*

- Preprocessed engagement data from 3,000+ students, resolving missing values and standardizing predictors across multiple time windows.
- Built and validated logistic regression models to assess the relationship between Quercus engagement and dropout risk, identifying early disengagement in first 4 weeks as a critical retention indicator.
- Delivered stakeholder-ready reports with clear visualizations and accessible interpretations, informing refinement of current learning support strategies.

Sep 2024 – Apr 2025 **Fair Machine Learning in Healthcare**, *STA496*

- Developed R/Python data pipeline (dplyr, Pandas, PyTorch) on 40,000+ EHR records from MIMIC-III database, producing a clean, standardized dataset that improved downstream model stability and interpretability.
- Executed PMI-SVD embedding-generation pipelines and performed KESER feature selection on 6,000+ clinical concepts for robust prediction tasks.
- Trained regression and random forest models in R for depression diagnosis and conducted fairness evaluation, uncovering demographic disparities driven by label bias.
- Built a fully reproducible data workflow and curated a GitHub repository, enabling seamless replication and improving research transparency.

May 2024 – Current **Aspect of Robust Regression Analysis**, *University of Toronto Department of Statistical Sciences*

- Developed and evaluated 4 parameter-estimation approaches for high-dimensional, heavy-tailed data, producing methodological insights that improved estimation accuracy over traditional regression methods.
- Implemented and optimized algorithms in R/Python, resolving non-convergence issues and increasing computational efficiency for large-scale simulations.
- Delivered a departmental research talk and released an R package with 20+ reusable functions, strengthening reproducibility and statistical communication.

Feb 2024 – Apr 2024 **Investigating The Factors Affecting Cycling Accident Severity: A Bayesian Hierarchical Mixtual Model Apporach**, *STA365*

- Conducted EDA on 800K+ accident records in Python (Pandas, NumPy, Matplotlib, Seaborn), identifying key temporal and environmental predictors of severe injuries.
- Engineered a Bayesian missing-data imputation pipeline (Scipy) using posterior predictive sampling, restoring 60K+ incomplete records with validated convergence diagnostics.
- Built a Bayesian hierarchical mixture model in PyMC incorporating time-of-day structure, spike-and-slab selection, and demographic mixture components, achieving measurable performance gains over baseline logistic regression.

Sep 2023 – Apr 2024 **An Agent-Based Simulation Approach to Investigate the Effect of Decreasing Birth Rates and the Efficacy of Potential Solutions**, *ROP399*

- Designed 20+ agent-based simulations modeling population dynamics and evaluating policy interventions for declining birth rates, identifying two high-impact levers that improved long-term population stability.
- Synthesized 30+ multidisciplinary sources into an annotated bibliography, strengthening model assumptions and improving simulation realism.
- Conducted time-series and regression analyses in Python (SciPy) to quantify intervention effects, revealing short- vs long-term trade-offs that informed policy recommendations, resulting in a poster presentation.

Health Studies Course Projects

Feb 2024 – Apr 2024 **Effectiveness of Usage of Virtual Reality in Raising Awareness about Cannabis Misuse Issues among Teenages**, *HMB342*

- Designed a randomized controlled trial study comparing virtual reality–based education with traditional seminars to evaluate effectiveness in raising awareness of cannabis misuse among Canadian high school students.
- Developed evaluation metrics and statistical analysis plan (pre/post surveys, T-tests, knowledge retention, behavioral intentions) to measure intervention impact on awareness and future cannabis use.
- Applied health promotion and program design frameworks to integrate emerging technologies into adolescent substance misuse education, highlighting cost-effectiveness and policy implications for public health strategies.

Feb 2023 – Apr 2023 **Food Insecurity Intervention Program Planning**, *HST330*

- Conducted root cause analysis and developed a program proposal to address food insecurity among Toronto low-income families, modeling intervention design on the Supplemental Nutrition Assistance Program with a digital e-coupon system.
- Designed a quasi-experimental evaluation plan, including baseline and post-intervention measures (Food Insecurity Experience Scale, BMI, Healthy Eating Index, stress levels) to assess program effectiveness.
- Created a logic model and theory of change framework to outline expected outcomes, health implications, and long-term policy applications, demonstrating strong skills in program evaluation and health policy analysis.

Oct 2022 – Dec 2022 **Proposal to Address Teenager Substance Misuse Problem**, *HST209*
- Conducted comprehensive analysis of risk factors (family influence, peer pressure, media exposure, mental health) contributing to teenage substance misuse, synthesizing evidence-based research into actionable insights.
- Designed and evaluated multi-level interventions including individual counseling, family therapy, in-school workshops, and peer-led support communities to promote adolescent health and reduce substance misuse.
- Developed health promotion strategies grounded in empowerment and community engagement, aligning interventions with public health models to improve long-term outcomes for at-risk youth.

Other Experiences

Sep 2023 – Current **Teaching Assistant and Learning Support**, *University of Toronto Department of Statistical Sciences and Data Sciences Institute*
- Led tutorials and office hours across multiple statistics courses in person and online, strengthening undergraduate students and industry professionals understanding of R, Python, SQL, and statistical concepts.
- Supported high-school outreach events, Florence Nightingale Day, translating advanced statistical ideas into engaging, accessible explanations.

Sep 2022 – Apr 2023 **First-year Learning Community Assistant Peer Mentor**, *University of Toronto*
- Facilitated 12 sessions supporting 20+ first-year students in academic planning, study strategies, and transition management.
- Built an inclusive learning community and coordinated discussions with faculty and library staff to enhance student engagement.
- Managed the online Quercus course page to facilitate remote communication and resource access.

May 2022 – Sep 2022 **Bilingual Client Care Representative**, *Telus Health*
- Served as first point of contact for 100+ daily client inquiries, delivering bilingual (English/Chinese) support that improved satisfaction and reduced escalations.
- Assessed client needs, identified the purpose of each inquiry, and provided tailored service recommendations, ensuring clients felt informed, supported, and guided toward the right solutions.
- Coordinated service delivery by booking appointments and routing callers to the appropriate service-specific teams, contributing to faster resolution times and smoother operations.